

Hao Wu

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EDUCATION

Peking University

M.S., School of Software & Microelectronics

Sept. 2026 – Present

Beijing, China

Sichuan University

B.E., School of Cyber Science and Engineering

Sept. 2022 – Jun. 2026

Chengdu, China

- Rank: 1 / 161; GPA: 3.85 / 4.0

PUBLICATIONS & OPEN-SOURCE SOFTWARE

[†]Equal contribution. 3 additional manuscripts under review (2× NeurIPS, 1× ASOC).

Juntang Wang[†], Yihan Wang[†], Hao Wu[†], Jiayu Gao[†], Shixin Xu, and Dongmian Zou. “StageGuard: Physiologically Constrained Sleep Staging.” *KDD 2026*.

Juntang Wang[†], Hao Wu[†], Yihan Wang[†], Dongmian Zou, and Shixin Xu. “Mixing Configurations for Downstream Prediction.” *ICML 2026*.

Hao Wu, Jiangchuan Chen, Wengang Ma, Ping He, Xiaolong Lan, Tao Li, and Junjiang He. “LLM-Based Immune Detection Method for Unknown Network Attacks in ICS Under Few-Shot Conditions.” *ICIC 2025*.

Juntang Wang[†], Yihan Wang[†], Hao Wu[†], Dongmian Zou, and Shixin Xu. “Brain-Inspired Perspective on Configurations: Unsupervised Similarity and Early Cognition.” *BICS 2025*.

mheatmap | Proportional heatmap visualization and matrix analysis with label alignment | 682 Stars | [\[GitHub\]](#)

llmDataG | High-quality tabular data augmentation via LLMs | 40 Stars | [\[GitHub\]](#)

GCN4ETD | Encrypted traffic detection via graph convolutional networks | 11 Stars | [\[GitHub\]](#)

RESEARCH & WORK EXPERIENCE

Unsupervised Representation; Physics/Structure-Prior ML; Biomedical AI; Graph & Matrix Visualization

Dec. 2024 – Present

Duke Kunshan University, Division of Natural and Applied Sciences; Research Assistant

Kunshan, China

Advisors: Prof. Shixin Xu and Prof. Dongmian Zou

- **MixConfig (ICML 2026)**: A plug-and-play feature augmentation module for downstream prediction. It extracts stable clustering configurations at multiple resolutions and learns sample-specific mixing weights, allowing predictors to use structure at different scales.
- **StageGuard (KDD 2026)**: A backbone-agnostic framework for physiologically constrained sleep staging. It combines a soft transition penalty during training with duration-aware semi-Markov decoding to discourage implausible transitions and excessively short sleep-stage bouts.
- **DSNI**: A benchmark for predicting bacterial cultivation conditions from 16S rRNA sequence data.
- **MINN**: Manifold-prior learning for electrolyte inference from ECG signals.
- **PPIDM-II**: Physics-informed diffusion modeling with an abstention mechanism.
- **DriftNeRF**: Neural rendering from sparse views.
- **mheatmap**: An open-source tool for proportional heatmap visualization and matrix analysis, with label alignment for comparing clustering configurations.
- **Lambda Lens**: Visualization of graph-embedding parameters to support inspection of representation structure.
- **Robotic-arm thermal compensation**: An industry collaboration on modeling thermal drift and compensating for localization errors.
- **Adhesive-dispensing optimization**: An industry collaboration on process-quality prediction and parameter recommendation.

Computer Immunity; Network Intrusion Detection; Noisy-Label Learning

Jul. 2023 – Jun. 2026

Sichuan University, Cyber Range Innovation Research Center; Research Assistant

Chengdu, China

Advisors: Prof. Tao Li and Prof. Junjiang He

- Designed an antigen-adaptive evolution algorithm based on immune gene mechanisms, improving detector diversity and attack-feature coverage.
- For few-shot unknown-attack detection, designed an LLM-optimized artificial immune algorithm: prompt templates guide the model to analyze antigen-space distributions and iteratively generate mutant antibodies, with immune tolerance used to filter detectors.
- Tackled noisy labels in encrypted traffic via a multi-resolution community-structure method, using neighborhood consistency, community support, and cross-granularity stability to identify mislabeled samples, then encoded community configurations as structural features to improve downstream robustness.

RAG-based Question-Answering Service

Jun. 2025 – Jul. 2025

Topsec Networks (Beijing), Product R&D Intern

Chengdu, China

- Built a LangChain + Faiss RAG pipeline covering document chunking, vector indexing, similarity retrieval, and context-augmented generation.

HONORS & COMPETITIONS

- **Sichuan Province Outstanding Graduate**, 2026.
- **Sichuan Province Class-A Comprehensive Quality Certification**.
- **Ningxia Yanbao Scholarship**, 2025.
- **Sichuan University Outstanding Student Award**, 2023, 2024, 2025.
- **Sichuan University Second-Class Scholarship**, 2023, 2024, 2025.
- **National Project Leader**, National College Student Innovation & Entrepreneurship Program, 2024. Project: anomaly detection and source tracing for multi-source heterogeneous smart-grid data.
- **National Second Prize**, China Undergraduate Mathematical Contest in Modeling, 2023. Project: optimization modeling and solution using nonlinear programming.
- **Sichuan Second Prize**, China Undergraduate Mathematical Contest in Modeling, 2024. Project: bench-dragon motion analysis using the bisection method.
- **Sichuan Bronze Award**, 9th “Internet+” College Student Innovation & Entrepreneurship Competition.
- **Third Prize**, China Undergraduate Mathematics Competition, 2023.

ADDITIONAL INFORMATION

Service: Grade Leader of the Class of 2022, School of Cyber Science and Engineering, Sichuan University; President of St.808 Student Association; Lead in SCU’s “Youthful SCU” promotional music video.

Skills: Python, SQL, C; scikit-learn, PyTorch, NumPy, pandas; Git, $\LaTeX$.

Certification: Sangfor security certification (SCSA-S).